

Navneet Yadav, PhD

Materials Scientist · Former Royal Society Newton International Fellow

📍 Swansea University, United Kingdom

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About

Synthetic inorganic materials scientist and analytical chemist with over four years of post-PhD research experience developing **noble-metal and multimetallic nanomaterials** for **biomedical sensing**, **antimicrobial and anticancer applications**, and **electrochemical wastewater treatment**. Former Royal Society Newton International Fellow, experienced in translating fundamental materials concepts into experimentally validated solutions, with expertise in advanced characterisation, mechanistic interpretation, grant development, laboratory management and postgraduate mentoring.

Professional Experience

Research Associate

Jul 2026 – Present

Department of Chemistry, Swansea University, United Kingdom

- Synthesising magnetic nanoparticles and noble-metal nanostars for biomedical sensing, developing a nanosensor for early-stage diagnosis of pancreatic cancer.
- Using Raman, UV-Vis, XRD, XPS, TEM and SEM-EDS with elemental mapping to establish structure–property relationships.
- Contributing to experimental planning, grant writing, research documentation and day-to-day laboratory operations.

Royal Society Newton International Fellow

🏅 Royal Society

Feb 2023 – Jan 2025

Department of Chemical Engineering, Swansea University, United Kingdom

- Developed and optimised nanostructured anodes for electrochemical oxidation of emerging organic pollutants in wastewater.
- Linked material composition and surface properties with treatment performance using XRD, XPS, Raman and SEM-EDS/mapping.
- Investigated non-thermal cold plasma and ozonation for degradation of PFOA, azithromycin and erythromycin; used HPLC-MS for by-product identification and pathway analysis.
- Mentored postgraduate researchers in nanomaterial synthesis, characterisation, electrochemistry and advanced oxidation processes.

Research Associate

Jul 2021 – Jan 2023

University of Allahabad, India

- Synthesised multimetallic and transition-metal oxide nanomaterials by microwave-assisted green synthesis and chemical reduction.
- Performed XRD, XPS, Raman, TEM and SEM-EDS/mapping; evaluated antimicrobial and other biomedical applications.
- Supported grant writing, research-fund utilisation, undergraduate and postgraduate laboratories, and student research.

Education

PhD

2015 – 2021

University of Allahabad, India

Thesis: *Synthesis and Characterizations of Nanomaterials for Industrial Applications*. Microwave-assisted chemical-reduction synthesis of noble monometallic and multimetallic nanomaterials, their advanced characterisation (XRD, SEM-EDS, XPS, Raman, FTIR, DLS) and their antimicrobial and anticancer activity.

MSc

First Division

2012 – 2014

University of Allahabad, India

BSc

First Division

2009 – 2012

Faculty of Science, University of Allahabad, India

Publications & Patent

PATENT

A Composition for Microwave Assisted Green Synthesis of Trimetallic Au/Pt/Cu Nanoparticles Granted

N. Yadav, K. K. Dey, R. R. Yadav

Indian Patent 547594 (2024)

ENVIRONMENTAL REMEDIATION & ENERGY MATERIALS

1 **Electrochemical Performance of Titanium (Ti), Antimony-Doped Tin Oxide on Titanium (NiATOTi), and Platinum-Antimony-Doped Tin Oxide on Titanium (NiATOTiPt) Anodes for Wastewater Treatment**

N. Yadav, S. Younesi, I. P. Throlley, R. E. Palmer, C. Tizaoui

Chemical Engineering Journal 530 (2026), 173281 · doi:10.1016/j.cej.2026.173281

2 **High-Entropy Oxide ($CaCu_3Ti_{3.85}Mn_{0.05}Sn_{0.1}O_{12}$) as a Multifunctional Electrocatalyst for Supercapattery and Water Electrolysis**

Y. Singh, P. Mishra, **N. Yadav**, N. K. Singh, V. Kumar, M. A. Reddy, B. Kumar

Nanoscale 18 (2026) · doi:10.1039/D6NR00812G

3 **Highly sensitive humidity sensor based on a composite of MoS_2 and multi-walled carbon nanotube sensing film**

S. Yadav, P. Chaudhary, **N. Yadav**, B. C. Yadav

Physica Scripta 100 (2025), 055945 · doi:10.1088/1402-4896/adc8e7

4 **Facile synthesis of the trimetallic Au-Pt-Pd nanofluids and performance for methylene blue degradation**

A. K. Maddheshiya, **N. Yadav**, S. Yadav, P. S. Yadav, T. P. Yadav

Materials Letters 349 (2023), 134735 · doi:10.1016/j.matlet.2023.134735

5 **Non-functionalized Au nanoparticles can act as high performing humidity sensor**

N. Yadav, P. Chaudhary, K. K. Dey, S. Yadav, B. C. Yadav, R. R. Yadav

Journal of Materials Science: Materials in Electronics 31 (2020), 17843–17854 · doi:10.1007/s10854-020-04338-y

BIOMEDICAL APPLICATIONS

6 **Synergistic Multifunctional Biomedical Activity of Microwave-Synthesized Au/Pt/Cu Trimetallic Nanofluids: Experimental and DFT Insights**

N. Yadav, A. Rafi, L. Kumar, B. Kumar, S. Paswan, A. K. Maddheshiya, J. Devi, N. Patel, A. Larimi, B. Kumar

RSC Advances (2026) · doi:10.1039/d6ra06054d

7 **Emerging trends and structure–activity insights of Schiff base–lanthanide complexes as antibacterial agents** Review article

S. Paswan, N. Jaiswal, Urvashi, M. Kumar, A. Nath, D. Mishra, **N. Yadav***

RSC Advances 16 (2026), 3478–3508 · doi:10.1039/D5RA08181E

8 **Synthesis and characterization of chemically functionalized novel smart guar gum xanthate based hydrogel: swelling, isotherm, kinetics, thermodynamic and reusability studies**

A. Chaurasiya, P. P. Pande, R. Shankar, P. Khare, P. Kumar, **N. Yadav**, K. K. Dey

Journal of Physics and Chemistry of Solids 200 (2025), 112584 · doi:10.1016/j.jpcs.2025.112584

9 **Synergistic Activity of Noble Trimetallic Nanofluids: Unveiling Unprecedented Antimicrobial Potential and Computational Insights**

A. K. Maddheshiya, **N. Yadav**, M. Kumar, P. S. Yadav, T. P. Yadav, A. Dubey

ACS Applied Bio Materials 7 (2024), 5906–5924 · doi:10.1021/acsabm.3c01268

10 **Microwave assisted formation of trimetallic AuPtCu nanoparticles from bimetallic nano-islands: Why it is a superior new age biocidal agent compared to monometallic & bimetallic nanoparticles**

N. Yadav, R. R. Yadav, K. K. Dey

Journal of Alloys and Compounds 896 (2022), 163073 · find on Google Scholar

11 **Two birds with one stone: oyster mushroom mediated bimetallic Au-Pt nanoparticles for agro-waste management and anticancer activity**

V. K. Chaturvedi, **N. Yadav**, N. K. Rai, R. A. Bohara, S. N. Rai, L. Aleya, M. P. Singh

Environmental Science and Pollution Research 28 (2021), 13761–13775 · doi:10.1007/s11356-020-11435-2

12 **Pleurotus sajor-caju-mediated synthesis of silver and gold nanoparticles active against colon cancer cell lines: a new era of herbonanoceutics**

V. K. Chaturvedi, **N. Yadav**, N. K. Rai, N. H. Abd Ellah, R. A. Bohara, I. F. Rehan, N. Marraiki, G. E. Batiha, H. F. Hetta, M. P. Singh

Molecules 25 (2020), 3091 · doi:10.3390/molecules25133091

13 **Rapid eco-friendly synthesis, characterization, and cytotoxic study of trimetallic stable nanomedicine: A potential material for biomedical applications**

V. K. Chaturvedi, S. N. Rai, N. Tabassum, **N. Yadav**, V. Singh, R. A. Bohara, M. P. Singh

Biochemistry and Biophysics Reports 24 (2020), 100812 · doi:10.1016/j.bbrep.2020.100812

14 **Synthesis, thermal, photo-physical, and biological properties of mononuclear Yb^{3+}, Nd^{3+}, and Dy^{3+} complexes derived from Schiff base ligands**

S. Paswan, A. Anjum, **N. Yadav**, N. Jaiswal, R. K. P. Singh

Journal of Coordination Chemistry 73 (2020), 686–701 · doi:10.1080/00958972.2020.1741557

15 **Trimetallic Au/Pt/Ag based nanofluids for enhanced antibacterial applications**

N. Yadav, A. K. Jaiswal, K. K. Dey, V. B. Yadav, G. Nath, A. K. Srivastava, R. R. Yadav

Materials Chemistry and Physics 218 (2018), 10–17 · doi:10.1016/j.matchemphys.2018.07.016

MATERIALS PHYSICS & ALLOYS

16 **The emergence of Griffiths phase in $CaCu_3Ti_{4-x}Mn_xO_{12}$ (CCTMO, x = 1, 2 and 3) geometrically frustrated antiferromagnetic complex perovskite**

V. Kumar, A. Kumar, S. Singh, K. Kumar, M. K. Verma, V. S. Rai, G. Nirala, S. Upadhyay, **N. Yadav**, N. B. Singh, D. Tiwary, K. D. Mandal

Journal of Materials Science: Materials in Electronics 35 (2024), 136 · doi:10.1007/s10854-024-11930-z

17 **Mechanical, elastic and microstructural investigations on HCP phase high-entropy alloys**

A. K. Maddheshiya, **N. Yadav**, S. P. Singh, D. Singh, P. S. Yadav, R. R. Yadav

MAPAN 38 (2023), 1–8 · doi:10.1007/s12647-023-00674-6

18 **Probing into crystallography and morphology properties of MoS_2 nanoflowers synthesized via temperature dependent hydrothermal method**

N. Kumar, P. Siroha, H. Shankar, D. Singh, Y. Sharma, R. Kumar, Ramovatar, **N. Yadav**, K. K. Dey

Nano Express 3 (2022), 035001 · doi:10.1088/2632-959X/ac7cfe

19 **Study of ultrasonic attenuation and thermal conduction in bimetallic Au/Pt nanofluids**

A. K. Verma, **N. Yadav***, S. P. Singh, K. K. Dey, D. Singh, R. R. Yadav

Johnson Matthey Technology Review 65 (2021), 556–567 · doi:10.1595/205651321X16038755164270

20 **Mechanical and thermophysical properties of high-temperature Ir_xRe_{1-x} alloys**

N. Yadav, S. P. Singh, A. K. Maddheshiya, P. K. Yadawa, R. R. Yadav

Phase Transitions 93 (2020), 883–894 · doi:10.1080/01411594.2020.1813290

UNDER REVIEW

- Efficiency and mechanistic insights of cold plasma in the rapid degradation of azithromycin, *Environmental Science: Water Research & Technology*
- Investigation of degradation and mechanisms of perfluorooctanoic acid (PFOA) by atmospheric cold plasma, *Journal of Water Process Engineering*
- Rapid response of non-functionalized Ag nanoparticles as a humidity sensor
- Molecular engineering of julolidine-containing environment-responsive fluorophores: charge-transfer photophysics, coordination effects and biomedical function

Full list on Google Scholar.

Technical Expertise

SYNTHESIS & MATERIALS

Chemical reduction · Microwave-assisted green synthesis · Noble-metal & multimetallic nanomaterials · Transition-metal oxides · Ag@Au nanostars · High-entropy oxides · Inert-atmosphere handling

CHARACTERISATION

XPS · XRD · Raman · UV-Vis · TEM / HRTEM · SEM / FESEM-EDS & mapping · FTIR · DLS · Fluorescence · TPS · HPLC-MS

PROCESSES & APPLICATIONS

Electrochemical oxidation · Cold plasma · Ozonation / AOPs · PFAS & antibiotic degradation · Antimicrobial assays · Anticancer studies · Biomedical sensing

SOFTWARE

Origin · ChemSketch · ORCA (DFT) · Quantum ESPRESSO · MS Office

RESEARCH DELIVERY

Experimental design & optimisation · Structure–property analysis · Mechanistic interpretation · Grant writing · Manuscript preparation · Laboratory management · Research mentoring · Peer review

Awards & Fellowships

2022 **Royal Society Newton International Fellowship**, The Royal Society, London, UK (2022–25)

2018 **DST-SERB Senior Research Fellowship**, Department of Science and Technology, New Delhi, India (2018–20)

2015 **DST-SERB Junior Research Fellowship**, Department of Science and Technology, New Delhi, India (2015–18)

Training

2024 Scientific Entrepreneurship, The Royal Society & Imperial College Business School

2024 Effective Leadership, The Royal Society & Imperial College Business School

2024 Storytelling for Researchers, The Royal Society

2024 ECR Network Colloquium & Grant Writing, Learned Society of Wales

2023 XPS training, Henry Royce Institute, University of Manchester

2023 Introduction to Management, Imperial College Business School & The Royal Society

– HPLC-MS training, Agilent Technologies

Memberships & Service

RSC Member, Royal Society of Chemistry (MRSC)

ISE Member, International Society of Electrochemistry

MRSI Life member, Materials Research Society of India

RS Alumni member, The Royal Society & Newton International Fellowship Programme

✎ Peer reviewer for ACS, RSC, Elsevier, Wiley, Springer, MDPI, IOP and Bentham Science journals

🗣 English (advanced) · Hindi and Urdu (native)